



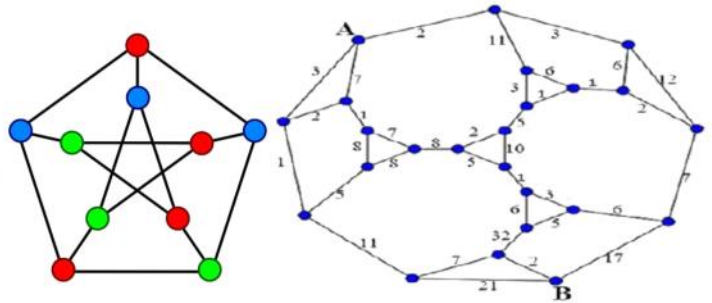
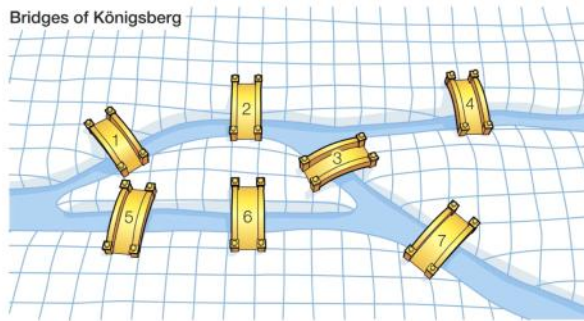
Course Code: 18B14CI645

Credit: 3

Dr Pankaj Dumka

Department of Mechanical Engineering

Bridges of Königsberg



Graph Theory: The Language of Connections

Fluery's Algorithm for Euler Trace and Circuit

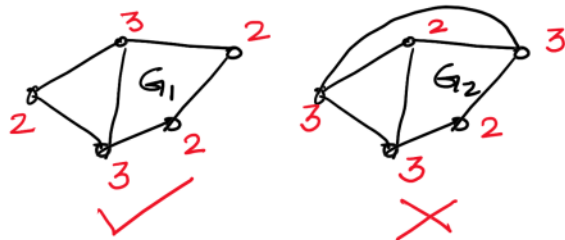
For highly connected graphs i.e. large no. of vertices
↓
manual observation is difficult

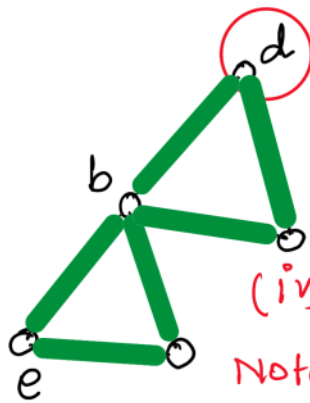
So, a systematic approach is required.

conditions in which Fleury's algo. is applied :-

- (i) The graph should be connected
- (ii) All vertices should have even deg. or exactly two vertices have odd deg.

which graph is a candidate for Fleury's algorithm





(i) create replica of graph

(ii) Start at any vertex.

(iii) Take an edge.

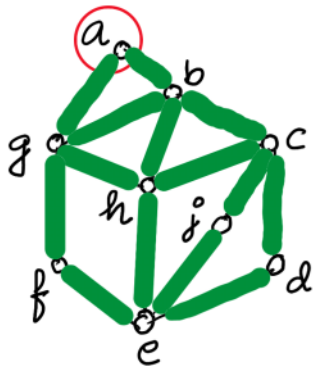
(iv) Delete that edge in replica.

Note:- If the deletion of any edge make the graph disconnected then take another edge.

$d-b-e-?-b-a-d$

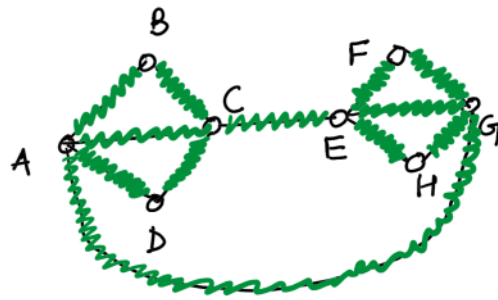
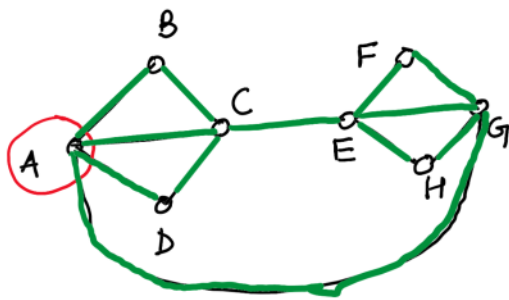
(v) If a vertex becomes isolated after edge deletion then, remove it also.

(vi) Repeat the process till the replica is no more visible.



$a - g - b - h - g - f - e - h - c - j - e - d - c - b - a$

↑
Euler circuit



$A-B-C-E-F-G-E-H-G-A-C-D-A$

Euler's circuit

Check all the deg's first.

use Fleury's algorithm to construct Euler trail or ckt.

